



*Software Engineering (3) Course (SWE521): Agile
Software Engineering Skills*

AGILE SOFTWARE ENGINEERING SKILLS

LECTURE 03 **Agile Roles**

2024 / 2025



Outline

1. Introduction
2. Scrum Master
3. Product Owner/On-site Customer
4. Product Owner Behaviors
5. Other Roles: For Larger Projects



Introduction

- **Self-organizing teams and how they work have been discussed.**
- In this chapter, we will focus on
 - roles outside the self-organizing team and **introduce** the skills you need to perform these roles.
 - The **scrum master** and **product owner** support the work of self-organizing teams.



Scrum Master 1 to 9

- **Scrum masters**
 - facilitate the scrum process on behalf of (instead of) the team,
 - monitor team status
 - and remove impediments (obstacles).
 - Practitioners consider the role as central to the success of the scrum method.
- Research that investigates what scrum masters actually do identifies **five** coordination activities:
 - process anchor,
 - stand-up facilitator,
 - impediment remover,
 - sprint planner
 - integration anchor.



Scrum Master 2 to 9 (process anchor)

Process Anchor

- The process anchor **mentors** team members in scrum method use. The agile process is described later.
- The idea is to **nurture** (care), **encourage** and perhaps gently **cajole** (convince) people to **learn** and **use** agile methods appropriately.
- This means understanding how agile methods work, by creating transparency about who is doing what.
- The scrum master **is not** a **team leader** or **supervisor** in the conventional sense.
- The scrum master does not tell people what to do.



Scrum Master 3 to 9 (process anchor)

• Process Anchor

- the scrum master has to **encourage**, sometimes recalcitrant (troublemaker) team members, to give their productive best.
- Agile teams often like to get into a **rhythm**, known as a *cadence*.
 - That means everyone **knows** what to **expect** when a **week starts** with sprint planning or **ends with** a customer demonstration.
- People can plan their **work**, and perhaps even their **social lives**, around this iteration cadence.



Scrum Master 4 to 9 (Iteration Planner)

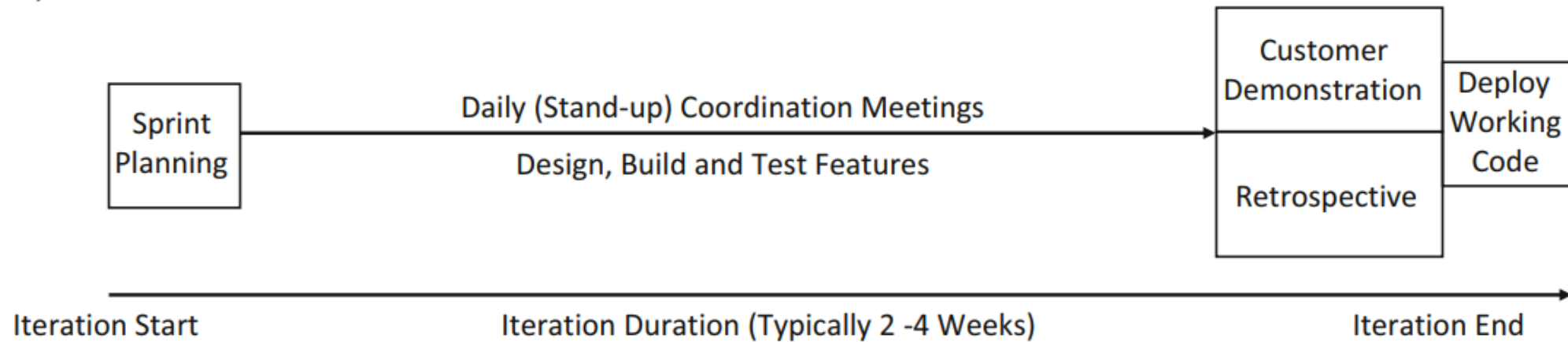
- **Iteration Planner**

- The **iteration planner** activity **helps select** and **estimate requirements** for implementation.
- Iteration planning is described later.
- **Everyone** on the **team** is **involved** in iteration planning for half a day, or a day, at the start of each iteration.
- The important point to **make** here is that the **scrum master facilitates** the iteration planning process.



Scrum Master 5 to 9 (Iteration Planner)

A)



B)



Fig. 3.1 (a) Iteration structure and (b) feature life cycle



Scrum Master 6 to 9 (Stand-up Facilitator)

• Stand-up Facilitator

- The **stand-up facilitator** conducts coordination meetings within a team.
- Facilitating a stand-up **involves** influencing the conduct of the meeting while saying as little as possible.
- When meeting participants **go off topic**, you **want to guide** them gently back to the purpose of the meeting.
- The whole **purpose** of the stand-up is to **discuss** and **disseminate** the status of the project.
- **Impediments** are **mentioned**, but **not discussed** in detail.



Scrum Master 7 to 9 (Stand-up Facilitator)

- **Stand-up Facilitator**

- Do not get drawn into trying to resolve impediments during the stand-up itself.
- You want to be alert for these and other issues that need to be discussed **outside** the stand-up meeting and **who** needs to be involved in those other conversations.
- It will be your responsibility to organize and facilitate those other meetings.
- Make sure you only involve those **who** need to be consulted.
- A common mistake is to let stand-up meetings go on too long, discussing topics that are not of interests to all the people involved and (from their point of view) **wasting** their time.



Scrum Master 8 to 9 (Impediment Remover)

- **Impediment Remover**

- The **impediment remover** eliminates work blockages for team members.
- **Impediments** are often **varieties** of **missing information** or lacking knowledge.
- A **scrum master** needs to find out who has the **elusive (complexed) knowledge** and convey that to the **blocked team member**.
- This process of information gathering requirements a combination of networking **skills** (knowledge of **who** knows **what**) and **diplomacy** (to convince people too busy or self-regarding to **part** (give up) with information).



Scrum Master 9 to 9 (Integration Anchor)

• Integration Anchor

- The **integration anchor** facilitates amalgamation (integration) of software elements.
- The whole point of a software development team is to produce software.
 - Someone has to **glue** together all the **bits** of software being created by the members of the team.
- From a feature perspective, each **piece** of code goes through a **plan**, **design**, **build** and **test** lifecycle as shown in Fig. 3.1b.
- The scrum master **needs** to **make sure** this code **integration happens**.
- This might involve checking up on someone who has taken on the task regarding the integration issue.
- You can **learn** more about some software tools you can use to help with this in Part III



Product Owner/On-site Customer 1 to 8

- The central **purpose** of product ownership is to communicate a **business need** to a development team.
- The **development team** **knows** how to **design**, **build**, **test** and **deploy** systems, and the **product owner** comes to know **what** **system needs** to be **built**.
- **Extreme programming** advocates an on-site customer, a **client representative** that is **available** to the team on a **full-time basis**.
- The product owner role is formally defined in scrum.
- **Product ownership** plays a **central role** in the overall software development process.



Product Owner/On-site Customer 2 to 8

- We now want to look at some of the activities product owners perform as part of their role.
- Taken together, these activities comprise the product owner role.
 - Product Grooming (Preparation)
 - Prioritizer
 - Release Master
 - Communicator
 - Traveller
 - Intermediary



Product Owner/On-site Customer 3 to 8 (Product Grooming)

- **Product Grooming**
- The **product owner** ensures that **requirements** bring **maximum value** to the business.
- In each iteration, the **product owner** decides **which** requirements from the **product backlog** are going to be most important for implementation in the **next iteration**.
 - Sometimes, this involves choosing to **prioritize** the **needs** of one customer group or segment over another.
- **Product owners** become experienced in **assessing** and **prioritizing** the **needs** of different segments of the customer base.



Product Owner/On-site Customer 4 to 8 (Release Master)

- **Release Master**

- In the release master activity, the product owner manages release plans and approves software source code for release to customers.
- **Early iterations** may not have sufficient code to deploy; see Fig. 3.2.
- **Approving releases** requires a **decision** about the **quality of software** (is it good enough to give to customers?) and **the scope** of the software (is there enough useful functionality to give to customers?).



Product Owner/On-site Customer 5 to 8 (Release Master)

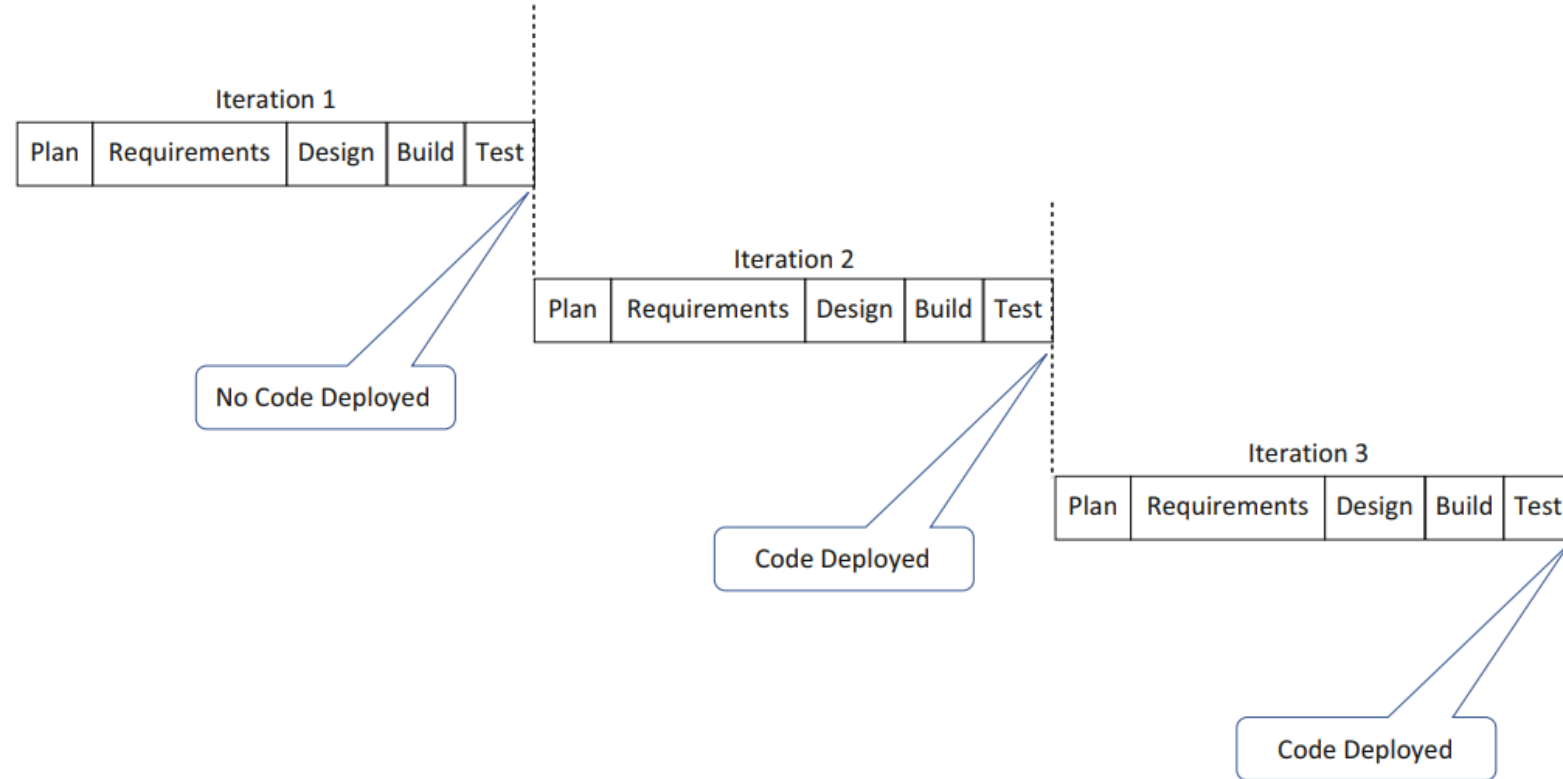


Fig. 3.2 Iterations and source code releases



Product Owner/On-site Customer 6 to 8 (Communicator)

- **Communicator**

- The **product owner** connects onshore (inside) and offshore (outside) **stakeholders** in the project team to manage geographical distribution.
 - Geographical distribution is not an ideal attribute for a project team.
 - We would prefer everyone to be located together in the **same site**.
 - The **ease of communication** and **movement** of digital goods means that geographical distribution has become a **feature** of software development programmes.
- The product owner, in the communicator activity, uses **audio** and **videoconferencing** and **online collaboration tools**.



Product Owner/On-site Customer 7 to 8(Traveller)

• Traveller

- In the traveller activity, the **product owner** spends time with geographically **remote stakeholders** gathering first-hand knowledge of their **needs** and **priorities**.
 - For example, a **product owner based offshore (outside)** will sometimes spend time (between 1 and 3 months, depending on the scale of the project) **on the client site** at the start of the project, becoming familiar with any **special features** of the **client's requirements**.
- The **traveller** is important for supporting development teams because they are based at the customer site and can **get answers to questions**.



Product Owner/On-site Customer 8 to 8 (Intermediary)

• Intermediary

- The product owner, in **the intermediary activity**,
 - **interfaces** (interactive) with **senior executives**,
 - driving software development programmes,
 - and disseminating domain knowledge to teams.
- **Domain knowledge** is understanding of the **business domain** or **sector** of the application software being created.
 - For example, it might be an application in travel, financial services or retail.
- To **perform the intermediary activities**, product owners **need** to have **extensive experience** of the particular system business domain.



Product Owner Behaviors 1 to 4

- There are **traits** (feature) that product owners **display** that are seen as desirable by their **line managers**.
- three main product owner behaviors we identified are to
 - favor face-to-face interactions,
 - understand and focus on real goals
 - make product owner teams well defined.



Product Owner Behaviors 2 to 4

- **Favor Face-to-Face Interactions**

- It is tempting to use digital technologies to overcome geographical distance.
- Written communication with email and word processing documents can help you deal with geographical, temporal (time) and cultural distances.
- However, understanding, trust and empathy come from building social capital through face-to-face interactions.
- It is very important to spend 'face-time' with stakeholders



Product Owner Behaviors 3 to 4

- **Understand and Focus on Real Goals**

- On successful projects, **product owners** appear to use **influencing skills** to keep a wide range of **stakeholders** targeted on a specific and focused **set of goals**.
- Creating a set of **clear, transparent** and **objective test criteria** allows **teams** to demonstrate progress toward **project goals**.
- **Product owners** **who** are able to stay focused on **key project goals** even as **inevitable challenges** and obstacles arise are **highly prized**.



Product Owner Behaviors 4 to 4

- **Make Product Owner Teams Well Defined**

- As has been suggested, for large projects,
 - the product sponsor,
 - intermediary,
 - technical architect
 - and other members

form a product owner team.

- The **process of building the product owner team** should be explicit and well defined.
- **Product sponsors** should create well-defined **processes** for product owner team building, induction of new members and succession planning.



Other Roles: For Larger Projects 1 to 4

- There are a couple of roles it is worth thinking about now:
 - product sponsor
 - and technical architect.



Other Roles: For Larger Projects 2 to 4

- **Product Sponsor**

- The **product sponsor** is a **project funder**.
- In business-to-consumer projects, each individual customer is a project funder.
- However, in a **business-to-business project**, or an internal project in a large organization, there is **an individual that signs off funding for a project**.
- Depending on the **size of organization**, this individual may be
 - a company **chief officer** (chief technical officer, chief financial officer or something)
 - or perhaps some other **senior executive**.



Other Roles: For Larger Projects 3 to 4

• Product Sponsor

- You may not see much of the project funder, if you have one.
- They may be **busy** with other responsibilities.
- The project funder is important. Ultimately, they are the persons that the team needs to satisfy.
 - So, listen carefully to what they say.
 - Try to understand the problem being solved, from their point of view.
 - Aim to see things from their perspective.
- That way, you stand a better chance of coming up with a solution that works for them.



Other Roles: For Larger Projects 4 to 4

- **Technical Architect**

- In smaller systems, **architecture is often straightforward**.
- You choose an appropriate architectural style and stick to it.
- You end up with a clear and simple high-level design.
- A design everyone can adopt, without much dissent.
- **On larger systems**, **high-level design or architecture** becomes much more sophisticated.
- Different subsystems may have contrasting architectural styles.
- The choice of architectural style may be more controversial (discussing) and less obvious.
- An experienced technology **leader** might be required to have the gravitas to carry the team towards a coherent approach.
- Hence, the overall system architecture can be much more finely balanced and complex.



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END OF LECTURE 03

Agile Roles

NEXT



Lecture Agile Roles (Exercises)